



Physiology

Variable expansin expression in *Arabidopsis* leads to different growth responsesHoe-Han Goh^{a,b}, Jennifer Sloan^a, Robert Malinowski^{a,c}, Andrew Fleming^{a,*}^a Department of Animal and Plant Sciences, University of Sheffield, Sheffield S10 2TN, United Kingdom^b Institute of Systems Biology, Universiti Kebangsaan Malaysia, UKM Bangi 43600, Selangor Darul Ehsan, Malaysia^c Laboratory of Plant Molecular Biology, Polish Academy of Sciences Botanical Garden – Centre for Biodiversity Protection in Powsin, ul Prawdziwka 2, Warsaw 02-973, Poland

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ABSTRACT

Expansins have long been implicated in the control of cell wall extensibility. However, despite ample evidence supporting a role for these proteins in the endogenous mechanism of plant growth, there are also examples in the literature where the outcome of altered expansin gene expression is difficult to reconcile with a simplistic causal linkage to growth promotion. To investigate this problem, we report on the analysis of transgenic *Arabidopsis* plants in which a heterologous cucumber expansin can be inducibly overexpressed. Our results indicate that the effects of expansin expression on growth depend on the degree of induction of expansin expression and the developmental pattern of organ growth. They support the role of expansin in directional cell expansion. They are also consistent with the idea that excess expansin might itself impede normal activities of cell wall modifications, culminating in both growth promotion and repression depending on the degree of expression.

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Introduction

Expansins are plant cell wall proteins which have long been implicated in the control of cell wall extensibility and, thus, in the control of plant growth (Cosgrove, 2005). Leading on from the initial purification and characterization of expansins in cucumber hypocotyl (McQueen-Mason et al., 1992), many authors have shown that expansins are abundant in rapidly growing tissue/organs. These include rice internodes (Cho and Kende, 1997; Lee and Kende, 2001), rice root hairs (Yu et al., 2011), oat coleoptiles (Cosgrove and Li Zhen, 1993), cotton fibers (Orford and Timmis, 1998), and elongating roots in soybean (Lee et al., 2003) and maize (Kam et al., 2005). Expansins are also involved in organogenesis, including leaf initiation in tomato (Reinhardt et al., 1998), and root hair initiation in pine (Hutchison et al., 1999) and *Arabidopsis* (Cho and Cosgrove, 2002). Furthermore, expansins play a role in non-growing tissues, often involving elements of cell wall differentiation. This includes ripening in various fruits (Brummell et al., 1999b; Harrison et al., 2001; Hiwasa et al., 2003; Sane et al., 2005; Yoo et al., 2003), differentiation of xylem cells in zinnia (Milioni

et al., 2001) and hybrid aspen (Gray-Mitsumune et al., 2004), and in poplar wood-forming tissue (Sterky et al., 1998).

In addition to correlative data, a number of papers have reported on the manipulation of expansin gene expression using transgenic methods in rice (Choi et al., 2003), tobacco (Pien et al., 2001; Sloan et al., 2009; Wang et al., 2011), tomato (Brummell et al., 1999a; Caderas et al., 2000), petunia (Zenoni et al., 2004, 2011) and *Arabidopsis* (Cho and Cosgrove, 2000; Goh et al., 2012). For example, the overexpression of endogenous expansin genes has been associated with increased coleoptile and mesocotyl growth in rice (Choi et al., 2003), and extended leaf petioles in *Arabidopsis* (Cho and Cosgrove, 2000). Although some of these reports have provided data essentially supporting a role for expansin in growth, it is noticeable that at least some reports suggest that a simple linkage between expansin gene expression and growth is not valid. For instance, the correlation found in specific up-regulated expansin gene expression association with leaf expansion in maize (Muller et al., 2007) was not found in the case of fescue leaves (Reidy et al., 2001) or tomato hypocotyls (Caderas et al., 2000). Indeed, the overexpression of an expansin isoform in tomato led to a repression in hypocotyl growth (Rochange et al., 2001). Thus, the precise relationship of expansins and plant growth remains unclear or at least variable.

Previously we reported on a series of experiments in which inducible overexpression of a cucumber alpha-expansin (EXPA) gene in tobacco led to an increased growth of the leaves, but only during a specific window of development (Sloan et al., 2009). These

Abbreviations: DAS, day after sowing; Dex, dexamethasone; DMSO, dimethyl sulfoxide; EXPA, alpha-expansin; GUS, β-glucuronidase.

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data supported the hypothesis that final leaf size is correlated with the maximal rate of leaf expansion (Cookson et al., 2005), which occurs during the phase of maximal growth sensitivity to induced expansin expression. This has also been substantiated by recently published results showing that the effects of suppressed multiple endogenous expansin gene expression depended on leaf developmental stage, with different responses between the leaf blade and petiole (Goh et al., 2012). We postulated that this differential sensitivity to expansin activity during development might account for some of the disparate data in the literature.

To further investigate this problem, we have generated a series of transgenic *Arabidopsis* plants in which the well-characterized heterologous cucumber EXPA gene can be inducibly overexpressed in *Arabidopsis*. Our data indicate that although (as in tobacco) induced expansin expression can lead to an increase in lamina area, there was a more dramatic effect of decreased petiole growth following high-level induction of expansin expression. This suppression of extension growth was also observed in the hypocotyls of these plants. These data indicate that the outcome of altered expansin gene expression is context-dependent, both in terms of organ and perhaps growth habit. They are consistent with the hypothesis that expansin acts to unlock elements within the cell wall to allow access and action of other cell wall components and that it is the nature of these co-acting molecules that determines the final phenotype observed.

Materials and methods

Plant growth conditions and treatments

Arabidopsis thaliana (L.) Heynh. (ecotype Col-0 from Nottingham Arabidopsis Stock Centre) seeds were surface-sterilized and stratified in the dark at 4°C for 1 week before being sown aseptically on solid media of 0.5× MS salts (Murashige and Skoog, 1962) (Sigma–Aldrich), 1% (w/v) sucrose, and 0.8% (w/v) plant agar (Duchefa Biochemie) in 12 cm × 12 cm square petri dishes (Greiner Bio One) with 36 seeds/plate. Growth conditions were 22°C day/20°C night under a 16-h photoperiod, light intensity of 100 μmol m⁻² s⁻¹.

For the induction of pOpON::CsEXPA1 transformant plants, growth media were supplemented with various concentrations of dexamethasone (Dex) prepared from 10 mM stock in dimethyl sulfoxide (DMSO) diluted using distilled water. Control media were supplemented with an equivalent concentration of DMSO (0.1% (v/v)). For staged-transfer experiment of gene expression study, plants grown on solid media at a density of 4 cm² per seed were categorized according to the number of the number of visible leaves (>1 mm width) and the size of leaf 5. Plants of equivalent developmental stage at 14 days after sowing (DAS) were chosen to be transferred to new small round petri dishes (5 cm diameter, *n* = 3 each) containing solid media supplemented with various concentrations of Dex for 24 h induction. For time-course study of dark-grown hypocotyl and root growth, seeds were grown at a density of 70 seeds per plate over three rows on 0.5× MS agar medium supplemented with 10 μM Dex or DMSO 0.1%. The petri dishes were double wrapped in aluminum foil.

Histochemical β-glucuronidase (GUS) assays of induced plants/leaves/hypocotyls were performed according to the standard protocols (Jefferson et al., 1986).

Generation of pOpON::CsEXPA1 transformant plants

The coding sequence CsEXPA1 (992 bp) from cucumber (Cuca.143960.1) was cloned into a Dex-inducible gene expression system under the control of bidirectional pOp6 promoter through TOPO cloning to generate transgenic *Arabidopsis*

(pOpON::CsEXPA1) of Columbia-0 (Col-0) background. TOPO cloning was performed according to the manufacturer's protocol (Invitrogen) and the product was used for heat transformation of DH5α chemically-competent *Escherichia coli* (Bioline). Inserts from pENTR/D-TOPO were recombined using LR clonase II (Invitrogen) into a Gateway-compatible binary vector pOpON2.1 (Wielopolska et al., 2005). Vector was transformed into *Agrobacterium tumefaciens* strain GV3101::pMP90RK through electroporation and then wild-type Col-0 *Arabidopsis* plants grown in Levingston M3 compost by the floral dip method (Clough and Bent, 1998). T1 progeny were self-pollinated, and subsequently selected using the appropriate antibiotic (50 μg/mL kanamycin), and genotyped using specific primers. Induction tests and histochemical GUS assays were performed in subsequent generations to select for T3 homozygous transformants with stable expression pattern upon induction. Single T3 homozygous transformant line 10/1 was used in this study.

Molecular characterization of pOpON::CsEXPA1 transformant plants

RNA extracted from whole-plant tissue using TRIZOL method (Chomczynski and Sacchi, 1987) were standardized to equal amount (4 μg) through spectrophotometric measurement and gel electrophoresis and used as template for the first-strand cDNA synthesis using M-MLV reverse transcriptase Rnase H minus (Promega) with oligo dT(18). Products were used as template for subsequent PCR amplifications. Semi-quantitative RT-PCR analysis of pOpON::CsEXPA1 used forward primer 5'-TTTGCTCTCACCTTCGCTGA-3' and reverse primer 5'-GCCTTGGCCATTGAGATAGT-3' for CsEXPA1, with BIOTAQ DNA polymerase (Bioline): initial denaturation at 94°C for 5 min, cycles: 15 s at 94°C, 30 s at 60°C, 1 min at 72°C, and 1 min final extension at 72°C. The PCR cycle during the exponential phase was paused at defined intervals at the end of extension and aliquots were routinely taken at cycle 24, 28, 30 and 32. All PCR reactions were performed with RBCS (At5g38410) forward primer 5'-CTTCCTCTATGCTCTCCTCC-3' and reverse primer 5'-GTTGTGCAATCCGATGATCC-3', and PP2A (At1g13320) forward primer 5'-ACGTGGCCAAAATGATGCAA-3' and reverse primer 5'-CGCCCCAACGAACAAATCACA-3' as internal controls. RT-qPCR experiments using SYBR® Green PCR master mix (Applied Biosystems) in 96-well optical reaction plates with optical adhesive covers (ABI PRISM™) were designed, performed and analyzed using a StepOnePlus™ Real-Time PCR system with its accompanying StepOne Software (Applied Biosystems, version 2.2). Primers were designed using QuantPrime (<http://www.quantprime.de/>). The PCR efficiencies of each primer pair (200–250 nM working concentrations) were determined from standard curve experiments using five 10-fold serial dilutions of cDNA to a final reaction volume of 20 μL. CsEXPA1 forward primer 5'-CCTCTCTCCAACATTTCGAC-3' and reverse primer 5'-TGGTACCCTACGAAAGGAGAC-3', and UBC21 (At5g25760) forward primer 5'-CTCTTAAGTGGGACTCAGGGAATC-3' and reverse primer 5'-TGTGCCATTGAATTGAACCTCTC-3' were used in comparative C_T experiments for quantitative analysis of CsEXPA1 gene expression level. All results were determined from three biological replicates, each with three technical replicates.

Protein analysis of pOpON::CsEXPA1 transformant plants

Cell wall protein extraction was performed as described previously (Rochange et al., 2001) using growing whole plant (21 DAS) from 10 μM Dex-treated and 0.1% (v/v) DMSO mock-treated wild-type and transgenic *Arabidopsis* plants. 10 μg of cell wall protein extracted from pooled samples of 36 plants (21 DAS) was loaded per well for immunodetection on protein blots. Proteins were separated in 12% SDS-PAGE gels then blotted onto

polyvinylidene difluoride membrane (Bio-Rad) and blocking with 7.5% (w/v) marvel milk in Tris-buffered saline (pH 7.3) containing 0.05% (v/v) Tween-20 for 1.5 h at RT. For immunodetection, 1:2000 dilution of primary antibody (CsExp1 rabbit antiserum) was used and hybridization was carried out for 1 h at room temperature. Membranes were washed five times (5 min) with Tris-buffered saline, 0.05% (v/v) Tween 20, then used for hybridization with 1:3000 dilution of secondary antibody (goat anti-rabbit antibody coupled with horseradish peroxidase) in 5% BSA for 45 min at RT. After washing, protein was detected using luminol as a substrate (ECL, Amersham, UK), 5 min film exposure.

Microscopy

For histology studies, leaves were fixed in ethanol/acetic acid (7:1 (v/v)), then hydrated in 70% (v/v) aqueous ethanol, cleared using 50% (v/v) commercial bleach and rehydrated in 70% (v/v) aqueous ethanol. Fixed leaves were mounted on slides with a drop of water and observed under Nomarski optics using a BX51 Olympus microscope fitted with a DP71 charge-coupled device (Olympus).

Growth measurements and analysis

For vegetative rosette growth analysis of pOpON::CsEXPA1 plants compared to Col-0 wild-type plants, non-destructive measurements were made using photographs taken using a digital camera (Sony Cybershot DSC-H3). For leaf measurement and analysis, dissected leaves were fixed and mounted on glass slides, unfolded whenever possible using a pair of blunt-end needles and flattened under cover slips. Photographs of arranged slides were taken using a digital camera (Sony Cybershot DSC-H3) on white background with a standard 15 cm ruler for scaling. Acquired photographs were processed using Photoshop (Adobe® version CS3) and ImageJ 1.44c (<http://rsbweb.nih.gov/ij/>) with manual hand-tracing of leaf outlines using an LCD tablet (Wacom DTI-520) to generate leaf silhouettes. For rosette area, a similar approach was taken by manually tracing the rosette outline, excluding the cauline leaves and separating the rosette with overlapping leaves to generate silhouettes of whole rosette for measurements. Similar approach was applied for hypocotyl and root growth analyses, except without the sample mounting procedure, and dark background (black cardboard paper) for hypocotyl. Photographs were taken under dim green light to avoid de-etiolation on day 4 and day 6. Hypocotyl and root lengths were measured with a calibrated scale bar by manual hand-tracing.

Absolute expansion/extension rate at time j (AER_j) over two time points ($j \pm 2$ or 3 d) was determined using the following equation: $AER_j = (m_j - m_{j-1}) / (t_j - t_{j-1})$, where m is the measurement in area or length, t is time, and m_j and m_{j-1} are measurements at times t_j and t_{j-1} , the previous time point. Relative expansion/extension rate at time j (RER_j) over two time points ($j \pm 2$ or 3 d) was determined using the following equation: $RER_j = \ln(m_j / m_{j-1}) / (t_j - t_{j-1})$, where m is the measurement in area or length, t is time, and m_j and m_{j-1} are measurements at times t_j and t_{j-1} , the previous time point. Leaf circularity is calculated as $4\pi \text{ area} / \text{perimeter}^2$. All statistical analyses were performed using Minitab® statistical software version 14.1.

Author contributions

Conceived and designed the experiments: HHG AF. Generate the transformant plants: RM. Protein analysis: JS. Performed the experiments: HHG. Analyzed the data: HHG AF. Wrote the paper: HHG AF.

Results

Generation and analysis of Arabidopsis plants with inducible expansin expression

To generate plants in which expansin could be inducibly over-expressed we cloned the cucumber CsEXPA1 sequence into the pOpON vector system (Moore et al., 2006). A number of independent transgenic lines were obtained and homozygous T3 lines acquired. RT-PCR (Fig. 1A) and qPCR (Fig. 1B) analyses indicated that the CsEXPA1 transcript was not detectable in the non-induced plants but that after induction with 0.1 μM or higher concentrations of dexamethasone (Dex), CsEXPA1 transcript was present in a dose-dependent manner, with a maximal level being obtained at a Dex concentration of 10 μM or higher. To investigate whether the presence of CsEXPA1 mRNA led to an elevated level of endogenous expansin protein we performed protein immunoblot analysis on Dex-induced seedlings (Fig. 1C) using an antibody which has been shown to detect a broad spectrum of EXPA proteins (Pien et al., 2001). These data indicated that the level of detectable expansin protein was increased in the Dex-induced pOpON::CsEXPA1 transgenic plants.

The pOpON vector contains a bidirectional promoter so that transcriptional activation of the cloned target gene also leads to transcription of a β -glucuronidase (GUS) reporter gene construct (Wielopolska et al., 2005). To analyze the pattern of expression of the CsEXPA1 gene following induction, we analyzed GUS activity in pOpON::CsEXPA1 plants after treatment with various concentrations of the Dex inducer. These results showed the expression of reporter gene throughout the plant after Dex treatment (Fig. 1D). A higher blue coloration was visible in the vascular tissue but some signal was present in essentially all tissues, including epidermis, mesophyll and vascular tissue (Fig. 1E). Focusing on the leaf, GUS signal first became faintly visible after treatment with Dex at 1 μM but only became obvious at 5 μM or higher (Fig. 1F). Surprisingly, the induction of expansin expression led to an overt suppression of plant growth which was apparent at Dex concentrations of 5 μM and higher (Fig. 1G). When induced plants were dissected, all parts of the plant seemed to show a growth repression, with leaf 3 onwards showing in addition a downwards curling (Fig. 1H). To quantify this response, we performed a series of growth analyses.

Analysis of rosette area growth in induced pOpON::CsEXPA1 plants

To investigate the growth suppression in pOpON::CsEXPA1 plants after induction of expansin gene expression we first analyzed the change in rosette area with time (Fig. 2). This gives a measure of the exposed leaf area of the plant with time and is a combination of both lamina area and the elongation of the leaf petiole which thrusts the lamina outward from the plant and, thus, influences self-shading by the leaves. As shown in Fig. 2A, there was no significant difference in the rosette area between wild-type (WT) and Dex (10 μM)-treated WT plants at any time point. However, when pOpON::CsEXPA1 plants were treated with 10 μM Dex or higher, a significant decrease ($p < 0.05$) in rosette area was measured 15 days after treatment. After induction with a relatively low (1 μM) Dex concentration there was a tendency for rosette area to increase relative to mock-treated plants, albeit not statistically significant. When the growth curves in Fig. 2A were expressed as absolute expansion rates (Fig. 2B) a similar pattern was observed. The absolute expansion rate of the rosette increased to a maximum level at approximately 15 days after sowing (DAS). Induction of the pOpON::CsEXPA1 plants with 10 μM Dex led to a significant ($p < 0.05$) decrease in expansion rate so that the maximum rate was lower than in control plants. This repression in expansion rate was

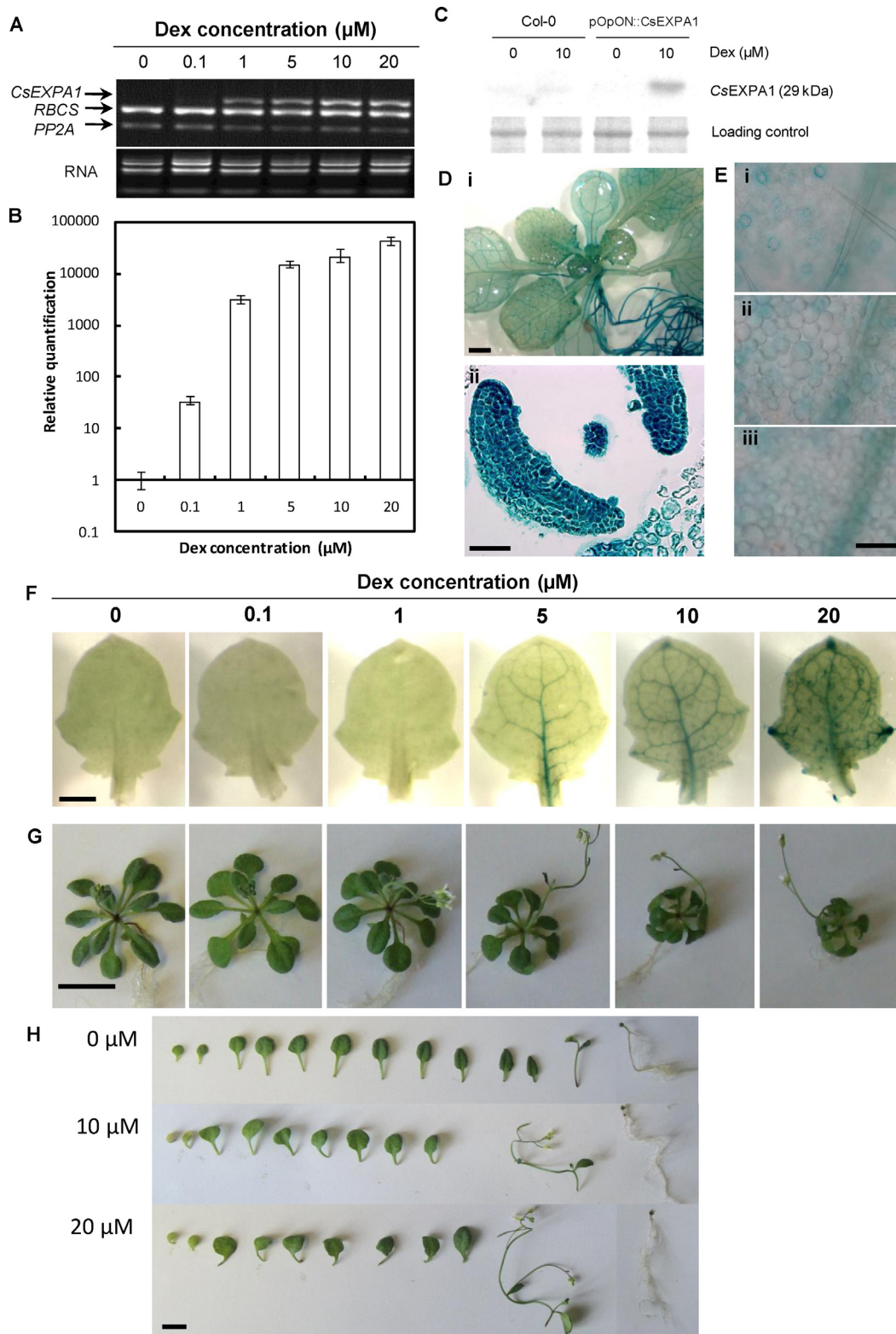


Fig. 1. Characterization of T3 homozygous pOpON::CsEXPA1 transformant plants. (A) Semi-quantitative RT-PCR of CsEXPA1 transcript abundance in 15 DAS plants after 24 h staged-transfer induction with different concentrations of Dex. RBCS and PP2A were used as internal controls, bands shown are at exponential cycle 30. (B) Quantitative real-time PCR (RT-qPCR) analysis of CsEXPA1 transcript abundance (as in A) based on $\Delta\Delta\text{Ct}$. UBC21 was used as internal control and 0 μM as reference sample, plotted on a Log-scale. Error bars represent minimum and maximum range of three biological replicates. (C) Protein immunoblot analysis of expansin expression (whole-plant tissue) from 21 DAS plants grown with or without 10 μM Dex. (D) GUS staining of an induced plant (i) and a magnified view of shoot apex cross-section (ii). (E) GUS staining showing induced expression across (i) epidermal, (ii) palisade, and (iii) vasculature layers of leaf 6. (F) GUS staining of leaf 6 harvested from 15 DAS plants after 24 h staged-transfer induction with different concentrations of Dex. (G) Plants grown on 0.5 $\times$ MS agar medium supplemented with different concentrations of Dex or DMSO 0.1% (v/v) (0 μM) 24 DAS. (H) Representative control (0 μM) and induced (10 and 20 μM) plants showing individual cotyledons, vegetative leaves, inflorescence and root. Bars: 1 mm in D and F, 50 μm in E, 10 mm in G, and 5 mm in H.

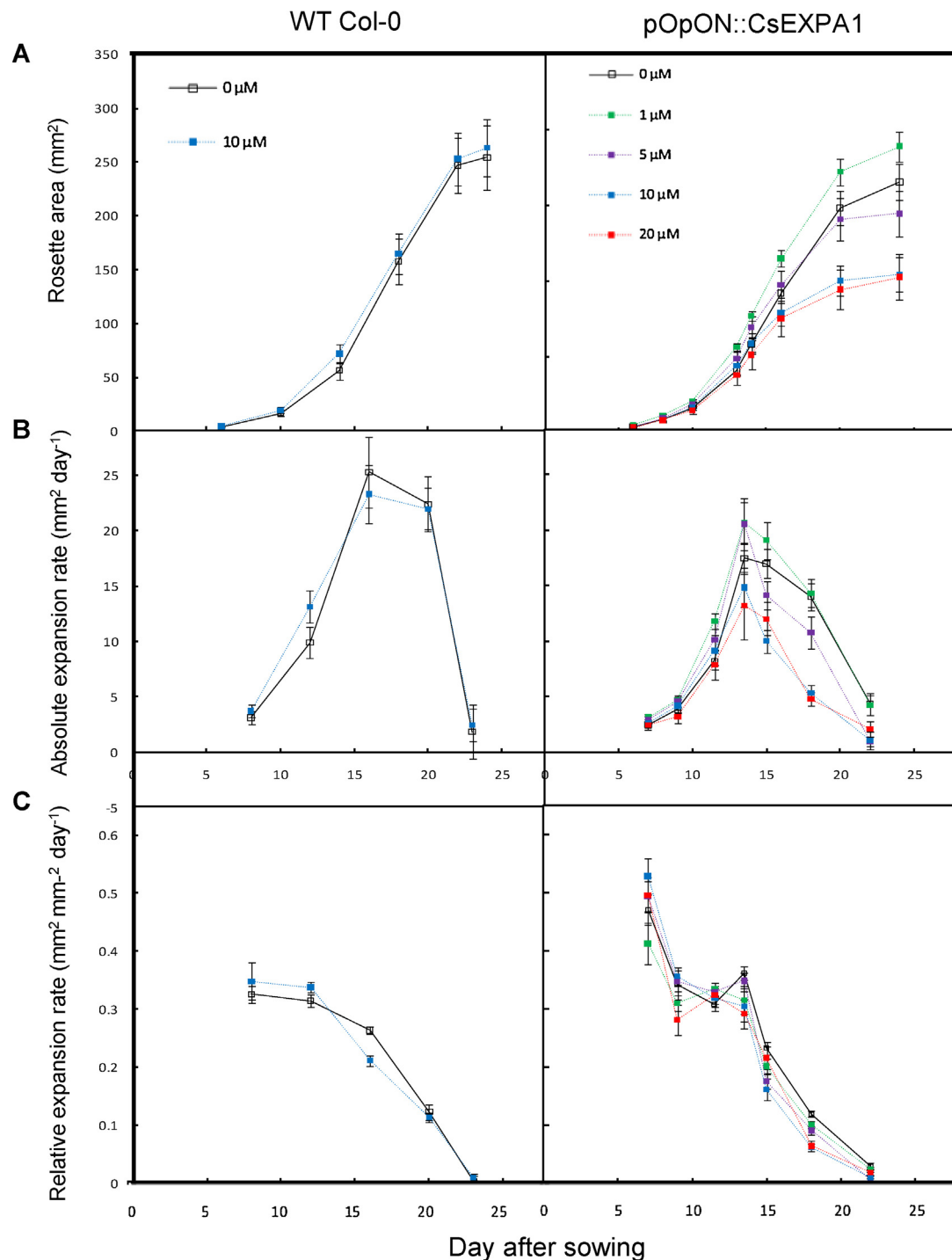


Fig. 2. Comparison between Col-0 wild-type and pOpON::CsEXPA1 vegetative rosette growth dynamics under different concentrations of Dex induction. (A) Rosette areas (coverage projections), (B) absolute rosette area expansion rates, and (C) relative rosette area expansion rates of 12 individual plants, grown on half-strength MS agar medium supplemented with different concentrations of Dex or DMSO 0.1%, were measured from 6 to 24 DAS. Values are means $\pm$ SE ($n = 12$).

maintained at time points beyond 15 DAS. In the 1 μ M Dex-induced pOpON::CsEXPA1 plants there was a tendency for an increased absolute rosette expansion rate relative to the control plants. This maximum occurred at a very similar time point in all treatments, i.e., the temporal pattern of rosette growth was not changed but the maximum growth rate achieved was altered. When expressed as relative rosette expansion rate (Fig. 2C) a similar general pattern was observed in all treatments, with the relative rate tending to decrease during rosette growth but with a slight hiatus at about 15 DAS (equivalent to the time point of maximum absolute growth

rate). Induction with higher Dex concentrations generally led to a lower relative growth rate, but the differences between treatments was less marked than for the data expressed as absolute growth rates (Fig. 2B and C).

To further investigate the growth responses measured at the level of rosette area, we examined the size parameters of the individual leaves which contributed to the rosette. For this analysis we focused on the growth of leaf 6 which previous work has established shows a typical development for a rosette leaf of *Arabidopsis* (Cookson et al., 2005; Kuwabara et al., 2011; Tsuge et al., 1996).

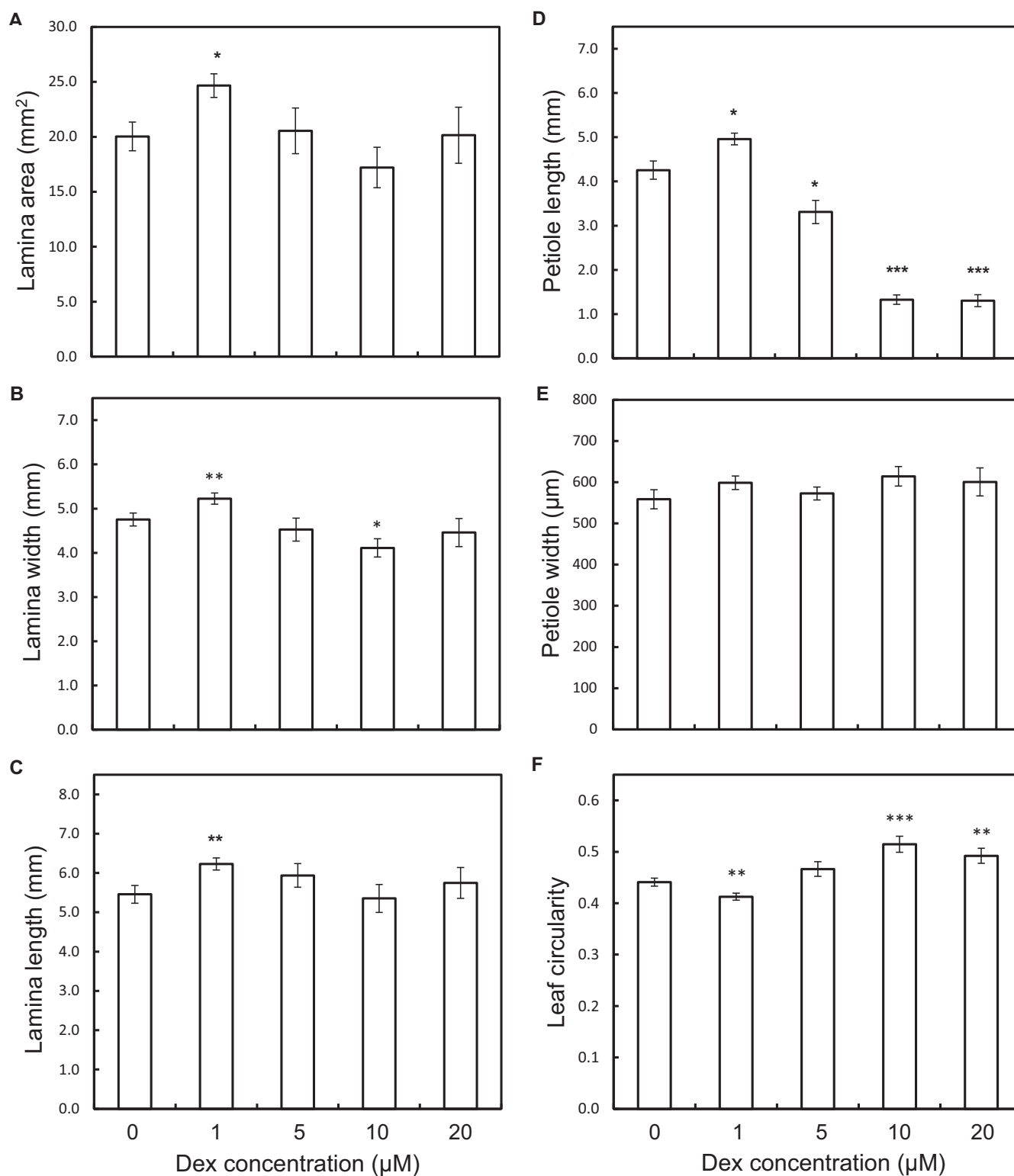


Fig. 3. Comparison of leaf 6 lamina and petiole parameters of 27 DAS pOpON::CsEXPA1 transformatant plants grown on medium supplemented with different concentrations of Dex. (A) Leaf lamina area. (B) Lamina width. (C) Lamina length. (D) Petiole length. (E) Petiole width. (F) Leaf circularity for morphometric comparison. Values are means $\pm$ SE ($n = 10$ –13 individual plants). One-way ANOVA, Tukey tests compared with 0 μ M control * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

As shown in Fig. 3A, when treated with a low level of the Dex inducer (1 μ M) there was a significant increase in final lamina area of the leaf. This was due to a significant increase in both lamina width (Fig. 3B) and length (Fig. 3C). At higher levels of Dex inducer there was no significant change in leaf lamina area, length or width (with the exception of width after 10 μ M Dex induction).

However, when petiole length was examined a different pattern of growth response was observed. Although after 1 μ M Dex induction a significant increase in petiole length was observed, at higher Dex inductions a massive repression of growth occurred (Fig. 3D). This growth repression was significant at the 0.1% confidence limit. Petiole width showed no significant change after Dex induction

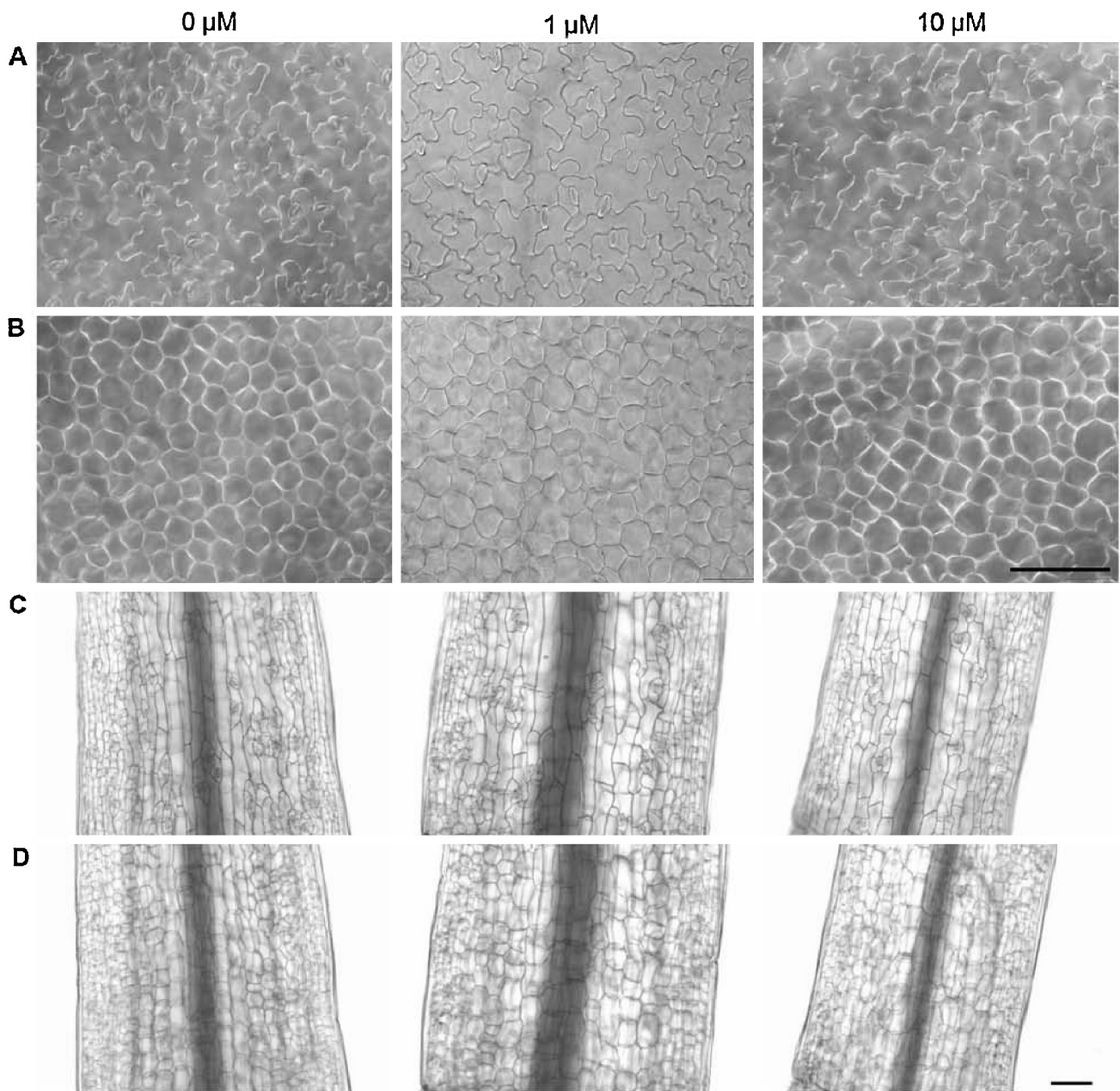


Fig. 4. Histological comparison of middle regions of leaf 6 lamina and petiole. Representative images from pOpON::CsEXPA1 transformant plants grown on 0.5× MS agar medium supplemented with DMSO 0.1%, 1 μM and 10 μM Dex, as shown. (A) Lamina adaxial epidermis. (B) Lamina mesophyll. (C) Petiole adaxial epidermis. (D) Petiole mesophyll. Bars = 100 μm.

(Fig. 3E). As a result of these altered parameters of leaf length, width and petiole length, the overall leaf shape was altered differentially depending on the level of pOpON::CsEXPA1 induction. It is this overall leaf shape (petiole + lamina) that contributes to the rosette growth described in Fig. 2. At lower levels of expansin induction the leaf showed a lower calculated value of circularity (was more elongated), whereas a higher degree induction of expansin induction led to leaves being significantly more compact (Fig. 3F).

Analysis of organ growth in induced pOpON::CsEXPA1 plants

Altered growth in a multicellular organ can occur either by a change in the number of cells generated during development (with each cell achieving a normal individual size), by a change in mean cell size (with no increase in the number of cells generated) or by a

combination of these processes. To investigate the cellular mechanism underpinning the growth responses described above, we analyzed cell size in the leaf lamina and petiole of leaf 6 in plants after induction with either a low (1 μM) or high (10 μM) concentration of Dex. A histological comparison of induced and non-induced leaf tissue is shown in Fig. 4. A statistical analysis of cell dimensions (Table 1) indicated that in the leaf lamina there was a tendency for an increase in epidermal and mesophyll cell area after induction with either 1 μM or 10 μM Dex. However, this was only significant ($p < 0.05$) in the epidermal cell layer after induction with 10 μM Dex, indicating that cell growth was affected differently in different tissue layers. As shown in Fig. 3, mean lamina area was not altered by this treatment, suggesting that the total number of cells in the epidermal layer was decreased following increased expansin gene expression, thus compensating for the increase in mean cell

Table 1

Histological analysis of cell size and cell number at the middle central region of leaf 6 lamina and petiole harvested on 27 DAS from pOpON::CsEXPA1 transformant plants induced with different concentrations of Dex since seed sowing.

Dex concentration (μM)	0	1	10
Lamina adaxial epidermis ($n = 10$)			
Pavement cell area (μm^2)	2326 $\pm$ 186	2552 $\pm$ 105	2797 $\pm$ 145
Cell number ^a	90 $\pm$ 4	84 $\pm$ 3	71 $\pm$ 5
Stomata number ^a	31 $\pm$ 1	28 $\pm$ 1	23 $\pm$ 2
Lamina mesophyll ($n = 28$)			
Cell area (μm^2)	544 $\pm$ 32	590 $\pm$ 36	657 $\pm$ 54
Cell width (μm)	27 $\pm$ 0.8	27 $\pm$ 0.9	30 $\pm$ 1.2
Cell length (μm)	27 $\pm$ 0.8	29 $\pm$ 1.0	30 $\pm$ 1.4
Cell number ^a			
Transverse	19 $\pm$ 1.0	19 $\pm$ 0.6	17 $\pm$ 0.3
Longitudinal	15 $\pm$ 0.7	14 $\pm$ 0.3	14 $\pm$ 0.6
Petiole mesophyll ($n = 20$)			
Cell area (μm^2)	666 $\pm$ 37	669 $\pm$ 46	873 $\pm$ 31
Cell width (μm)	25 $\pm$ 1.1	25 $\pm$ 1.0	31 $\pm$ 1.0
Cell length (μm)	35 $\pm$ 1.1	34 $\pm$ 1.7	36 $\pm$ 1.2
Cell number along cell file ^b	18 $\pm$ 1.0	20 $\pm$ 0.8	18 $\pm$ 0.8
Number of cell file ^c	20 $\pm$ 0.8	19 $\pm$ 0.4	16 $\pm$ 0.6
Petiole adaxial epidermis ($n > 10$)			
Cell length (μm)	187 $\pm$ 11	202 $\pm$ 4	157 $\pm$ 8
Number of cell file ^c	25 $\pm$ 0.6	26 $\pm$ 0.5	22 $\pm$ 0.6

Number of cells measured for lamina and petiole are stated as n . The values shown are mean $\pm$ SE of 6 biological replicates. Bolded numbers indicate statistical significance at $p < 0.05$ in 2-sample t -tests (cell size) or Mann–Whitney U tests (number of cell/cell file) compared with control.

^a Cell number was counted within the area captured by the micrograph (268 $\mu\text{m} \times 356 \mu\text{m}$).

^b Cell number was counted along the leaf-length direction on the third cell file from the margin of petiole (536 μm length).

^c The number of cell file was counted across petiole within the area captured by the micrograph (536 $\mu\text{m} \times 712 \mu\text{m}$).

size. Within the petiole, mesophyll cell area increased significantly ($p < 0.05$) after 10 μM Dex induction of expansin expression, whereas mesophyll cell size was unaltered after a lower level of induction. However, within the petiole epidermis the higher level of expansin induction led to a significant ($p < 0.05$) decrease in mean cell length, whereas the lower, 1 μM Dex induction led to a tendency for an increase in epidermal cell length. The number of cell files in the petiole was significantly ($p < 0.05$) decreased after the higher (10 μM Dex) induction of expansin expression.

Our analysis of the leaf growth response to elevated expansin gene expression indicated that relatively high levels of expansin induction led to growth suppression, particularly in the petiole. The petiole normally undergoes a relatively rapid uniaxial expansion relative to, for example, the leaf lamina where growth is more isotropic. To investigate whether this growth suppression in response to a high induction of expansin might be linked to the type or rate of growth, we examined the growth response of hypocotyls and roots of pOpON::CsEXPA1 seedlings (Fig. 5).

The hypocotyl normally undergoes rapid uniaxial expansion during germination and this growth occurs essentially without any accompanying cell division (Gendreau et al., 1997). The treatment of germinating pOpON::CsEXPA1 seedlings in dark with 10 μM Dex led to induction of transgene expression throughout the hypocotyl (Supp Fig. 1). RT-PCR analysis of these seedlings confirmed that the CsEXPA1 transcript was detectable only in the induced seedlings and that induction did not trigger down-regulation of its homolog counterpart in *Arabidopsis* (EXPA8) (Supp Fig. 1). Analysis of hypocotyl length over time indicated that induction of expansin gene expression in this tissue also resulted in a highly significant decrease in growth ($p < 0.001$) by day 6 after sowing (Fig. 5A). This was accompanied by a slight (but nevertheless significant) increase in hypocotyl width on 8 DAS.

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.jplph.2013.09.009>.

Conversely, when the growth of primary root length was measured after the induction of pOpON::CsEXPA1 seedlings, there was no significant difference in final growth relative to non-induced seedlings. Despite a tendency for root growth promotion after induction of expansin gene expression, the final root length was never significantly different from the control samples.

Discussion

In this paper, we report on the generation of transgenic *Arabidopsis* plants in which a previously characterized cucumber EXPA1 gene (McQueen-Mason et al., 1992; Pien et al., 2001; Rochange et al., 2001) can be inducibly expressed. Our analysis shows that following supply of the Dex-inducer there was a significant increase in CsEXPA1 mRNA and a clear increase in detectable expansin protein within the induced plants. Due to the nature of the construct used, the transcriptional activation of the CsEXPA1 transgene is linked to the transcriptional activation of the GUS reporter gene, which allowed a relatively simple histological assay of reporter gene activity to infer where CsEXPA1 gene expression was induced. Previous papers have used this system and have found a good qualitative correlation of target and reporter gene expression (Malinowski et al., 2011; Wielopolska et al., 2005). The results of this analysis indicated that following Dex-induction expression of CsEXPA1 occurred throughout the plant. Although a higher signal was apparent in the vascular tissue of the leaf, this probably reflects the histology of the tissue (relatively small, dense cells) compared to the surrounding tissue. Taken together, our data indicate that expression of the CsEXPA1 gene was negligible in the control, non-induced pOpON::CsEXPA1 plants but that following Dex-induction there was an accumulation of the CsEXPA1 protein in all parts of the plant. These plants allowed us to compare the response of *Arabidopsis* to elevated expansin gene expression with other plant systems in which various means have been used to alter expansin gene expression (Pien et al., 2001; Rochange et al., 2001; Zenoni et al., 2011).

Induction of CsEXPA1 led to a significant increase in lamina area at relatively low levels of inducer concentration. These results are consistent with data from other experimental systems showing that elevated expansin gene expression can lead to a promotion of leaf growth in tobacco and petunia (Pien et al., 2001; Sloan et al., 2009; Zenoni et al., 2011). However, a dramatic inhibition of petiole extension occurred when a higher level of induction was performed, with the growth suppression in this tissue following expansin expression overcoming the lamina phenotype, thus leading to an overall smaller plant. How should this apparently contradictory local growth response to increased expansin gene expression be interpreted?

Although a limited increase in expansin expression did lead to an increase in petiole and lamina growth, the *Arabidopsis* system was very sensitive to higher levels of expansin gene induction which led to growth suppression, i.e., the same effector can lead to opposite outputs depending on the degree of induction. It has been proposed that expansin acts to unlock the macromolecular structure of the cell wall and that it is the spectrum of other factors present in the cell wall which actually determines the resulting phenotype (Caderas et al., 2000; Rochange et al., 2001; Sloan et al., 2009). One possible explanation is that excess expansin could actually impede the access of other cell wall modifying proteins, such as XTHs to their sites of action within the cell wall. Sensitivity to excess expansin may also be linked to the intrinsic capability of the tissue for expansion since in both this study and others (Rochange et al., 2001), tissues which normally show a relatively

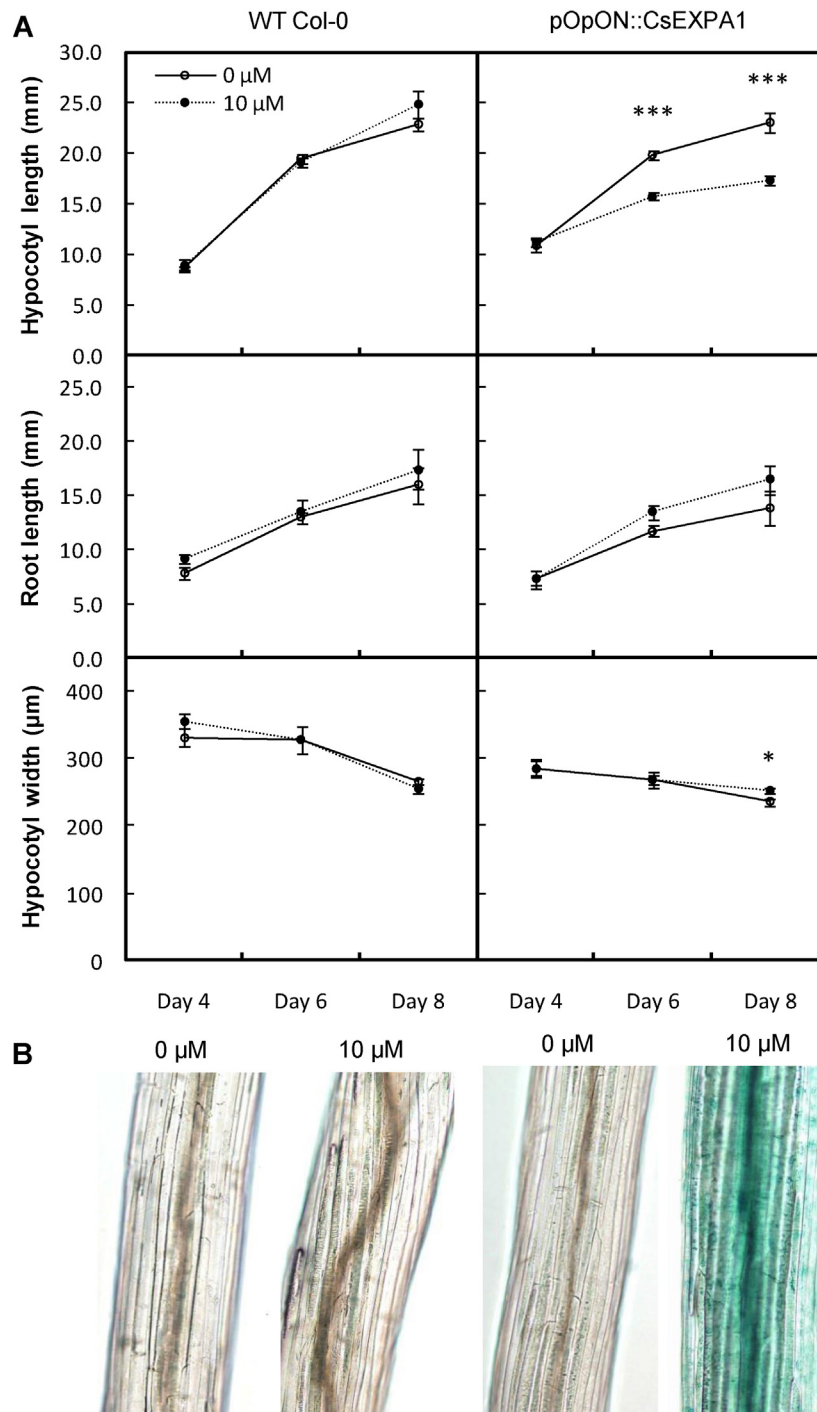


Fig. 5. Time-course study of hypocotyl and root growth of Col-0 wild-type and pOpON::CsEXPA1 transformant seedlings in the dark. (A) Comparison between hypocotyl and root length of seedlings induced with 10 μM Dex or mock-induced with DMSO 0.1%. Photographs were taken under dim green light on day 4 and day 6. Values are means $\pm$ SE ($n = 65$). Two-sample t -tests, * $p < 0.05$, *** $p < 0.001$. Hypocotyl widths were measured from micrographs at mid-point region between cotyledons and root boundary. (B) Representative middle regions of hypocotyl after GUS staining, with or without induction. Bar = 100 μm.

high rate of expansion (hypocotyls, petioles) showed a marked growth suppression following a high degree of expansin gene induction, which is not observed in slower, isotropically growing organs (leaf lamina). A better understanding of the actual molecular mechanism of expansin action and biochemical/structural characterization of the cell wall when expansin is in excess will be required to test these ideas.

A previous analysis of expansin function in *Arabidopsis* (Cho and Cosgrove, 2000) revealed an increase in petiole extension after

the expression of an endogenous expansin gene (*AtEXPA10*) via its promoter (which drives expression predominantly in the petiole). This increase in petiole length was accompanied by longer sub-epidermal cells. This suggests that in addition to the absolute level of expansin gene expression, local gradients might be important in determining the final growth outcome. This might be particularly important in the context of the epidermis/sub-epidermis since it has been proposed that the resistance to growth in these different layers might determine overall growth of an organ

(Savaldi-Goldstein et al., 2007) and there is certainly significant evidence of tissue-specific expression of many expansin genes (Cho and Cosgrove, 2002; Gray-Mitsumune et al., 2004; Lee et al., 2003; Lee and Choi, 2005; Reinhardt et al., 1998). In the experiments reported here, expansin expression was induced throughout all tissue layers, thus a uniform excess of expansin might obscure these endogenous gradients, again leading to alternative growth outcomes. Furthermore, cucumber expansin CsEXPA1 is closely related to *Arabidopsis* endogenous homologs AtEXPA2 and AtEXPA8 which were reported to be expressed in almost all plant organs (Goh et al., 2012). Therefore, the observed growth defects are unlikely to be due to the tissue-specific activity of cucumber expansin, rather they may reflect the nature of the cell walls in the growing organs. The observation that ectopic expression of CsEXPA1 led to no obvious effect on the transcript level of AtEXPA8 suggests there was no feedback to the regulation of endogenous expansin gene expression, but this requires further investigation of all of the expansin genes expressed in the target tissues reported here.

A final point to note is that while within the leaf lamina cell size tended to increase in response to induction of expansin gene expression (Table 1), this did not always lead to an increase in lamina area. It has been well documented that supra-cellular regulation can occur within a leaf, leading to compensation at the whole organ level for observed changes in component cell size (Tsukaya, 2002). The nature of this regulation is still obscure but it is possible that if expansin acts primarily at the cell level such supra-cellular regulators might counteract cell-based size control mechanisms. Within the petiole epidermis a high level of expansin gene expression was correlated with a significant decrease in cell length (which was not observed in the petiole mesophyll), suggesting that even different cell types within a single tissue can show quite distinct responses to an apparently similar alteration in expansin gene expression.

To conclude, the results reported here provide a description of the differential outcome of induced expansin gene expression on *Arabidopsis* growth. They fit with a theme in which the specific effects of ectopic expansin expression depend on the degree of induction of expansin expression and the endogenous developmental pattern of organ growth.

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